

Business Problem

A manufacturing and distribution company is facing delays in deliveries, inconsistent vendor performance, excess inventory in some warehouses, and frequent order processing issues. Management lacks a centralized dashboard to monitor the complete supply chain performance.

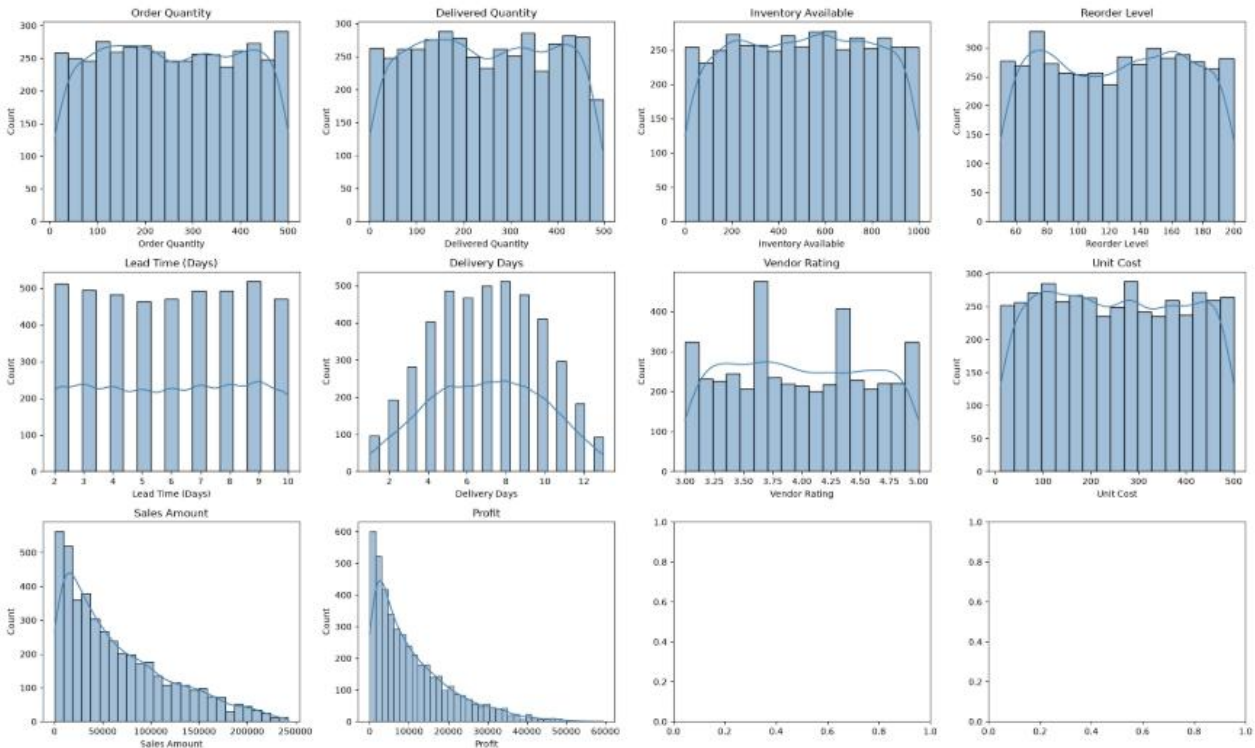
The goal of this analysis is to:

- Improve inventory visibility and stock management.
- Monitor on-time delivery performance across regions.
- Evaluate vendor performance and supplier reliability.
- Track order status for better operational control.
- Enable data-driven decision-making through interactive dashboards.

Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%	max	median
Order Quantity	4397.0	256.197180	142.045958	10.00	134.00	254.00	382.00	500.00	254.00
Delivered Quantity	4397.0	246.124176	142.178611	0.00	124.00	244.00	371.00	499.00	244.00
Inventory Available	4397.0	503.547191	286.424237	0.00	256.00	507.00	749.00	1000.00	507.00
Reorder Level	4397.0	125.041619	43.713512	50.00	86.00	126.00	163.00	200.00	126.00
Lead Time (Days)	4397.0	5.989766	2.598581	2.00	4.00	6.00	8.00	10.00	6.00
Delivery Days	4397.0	7.006595	2.924622	1.00	5.00	7.00	9.00	13.00	7.00
Vendor Rating	4397.0	3.987582	0.579856	3.00	3.50	4.00	4.50	5.00	4.00
Unit Cost	4397.0	253.878781	142.086379	10.00	129.00	253.00	377.00	500.00	253.00
Sales Amount	4397.0	65306.513759	55462.231938	156.00	19152.00	49455.00	98010.00	242550.00	49455.00
Profit	4397.0	10777.016259	9973.696333	16.79	2998.59	7722.83	15706.44	59497.87	7722.83



Order & Delivered Quantity: The average order quantity is about 256 units, while the average delivered quantity is about 246 units, showing a small delivery gap.

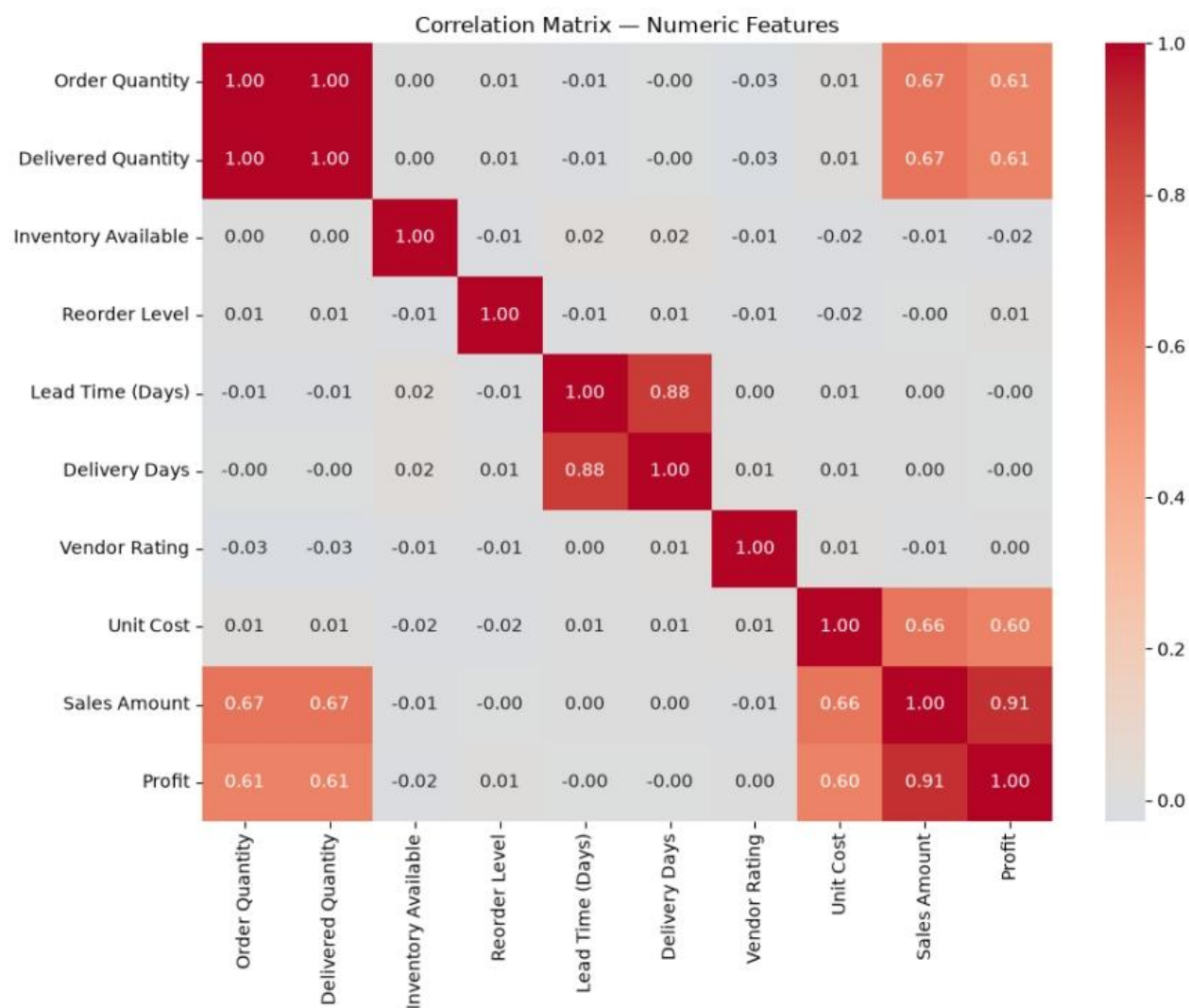
Inventory & Reorder Level: Average inventory is around 504 units compared with a reorder level of 125 units, indicating sufficient stock availability.

Lead & Delivery Time: Average lead time is approximately 6 days and delivery time is 7 days, suggesting relatively consistent delivery performance.

Sales & Profit: Average sales amount is around 65,307 with an average profit of approximately 10,777, indicating overall positive profitability.

Vendor & Cost Analysis: Vendor ratings average nearly 4/5, while the average unit cost is about 254, showing generally good vendor performance and stable costs.

Correlation Insights



Order & Delivered Quantity: Order Quantity and Delivered Quantity have a very strong positive correlation of 1.00, showing that deliveries closely match orders.

Lead & Delivery Time: Lead Time and Delivery Days show a strong correlation of 0.88, indicating that longer lead times are associated with longer delivery periods.

Sales & Profit: Sales Amount and Profit have a strong positive correlation of 0.91, meaning higher sales generally result in higher profits.

Unit Cost Relationship: Unit Cost has a moderate correlation with Sales Amount (0.66) and Profit (0.60), suggesting that product cost influences overall financial performance.

Overall Insight: Most inventory and vendor-related variables have weak correlations, while sales, profit, quantity, and delivery time show the strongest relationships

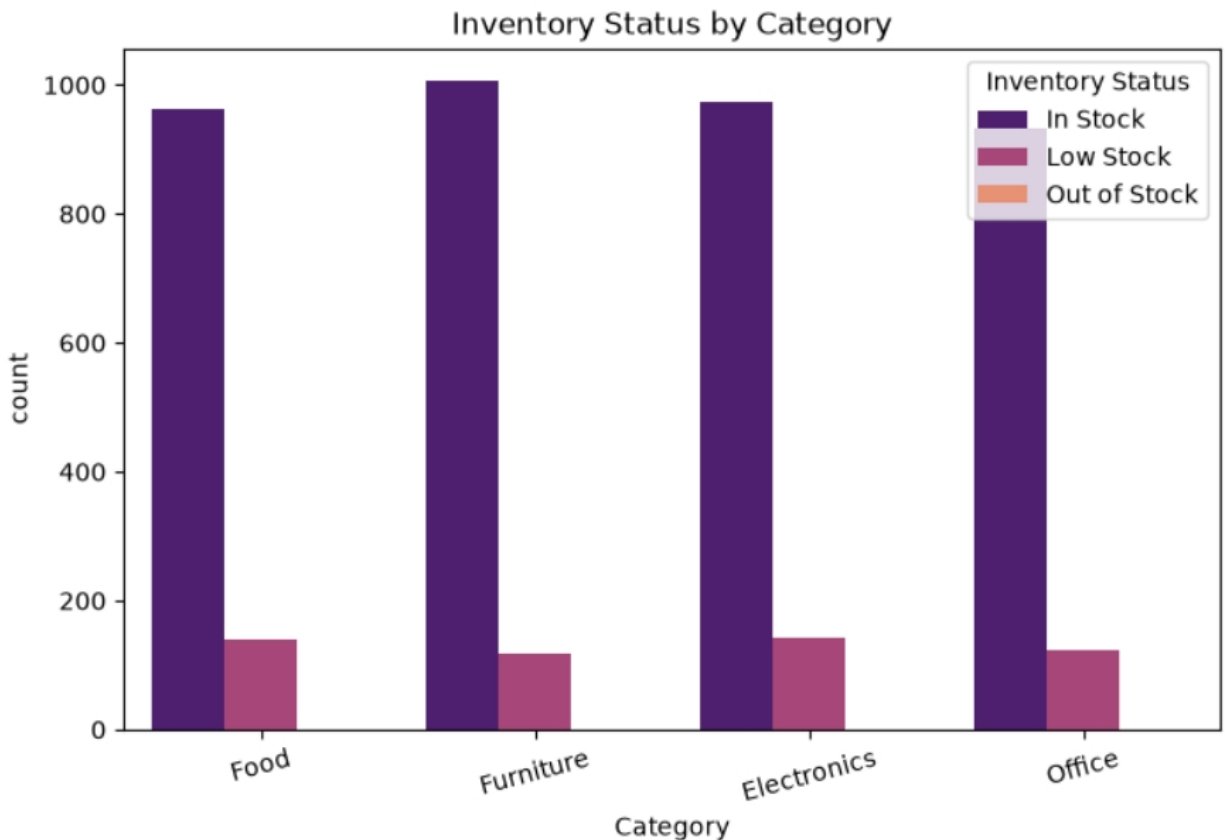
Research Questions & Key Findings

1. Total sales,Total profit,and profit margin by category

	Total_Sales	Total_Profit	Avg_Margin
Category			
Food	75617064	12554061.7	16.7
Electronics	74532422	12349951.2	16.6
Furniture	72073900	11767254.3	16.3
Office	64929355	10715273.2	16.5

- **Category-Level Sales & Profit:** Food leads with Rs 7.56 Cr in sales and Rs 1.26 Cr profit (16.7% margin), followed closely by Electronics (Rs 7.45 Cr, 16.6% margin).
- **Furniture & Office trail slightly:** Furniture posts Rs 7.21 Cr sales at 16.3% margin, while Office has the lowest sales (Rs 6.49 Cr) but a solid 16.5% margin.
- **Margins are tightly clustered (16.3%–16.7%) across all categories** — no category is significantly underperforming on profitability.
- Lower-selling categories still hold healthy margins, suggesting they could benefit from targeted marketing or pricing adjustments to boost volume without hurting profit.

2. Inventory Status by Category



- **In Stock dominates across all categories:** Furniture has the highest stock levels (~1000 units), followed by Food and Electronics (~950 units each), and Office slightly lower (~850 units).
- Low Stock is present in every category at a fairly consistent level (~100–150 units), with Food and Electronics showing marginally higher low-stock counts.
- Out of Stock is negligible across all categories — barely visible on the chart, meaning stockouts are rare overall.
- Stock health looks strong, but the consistent presence of Low Stock items in each category suggests ongoing replenishment monitoring is still needed to avoid future shortages.

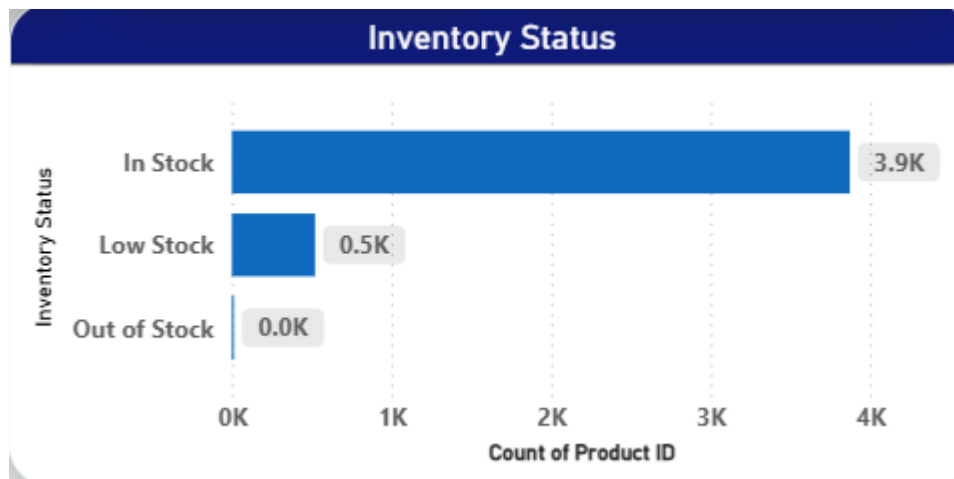
3. Vendor Performance

	Orders	Avg_Rating	Avg_Lead_Time	On_Time_Rate	Total_Sales	Total_Profit
Vendor Name						
Vendor 3	470	4.0	6.1	38.3	30889667	5091043.2
Vendor 8	460	4.0	6.1	39.6	30569458	5022115.4
Vendor 9	448	4.0	5.8	41.3	29850153	4934531.2
Vendor 1	442	4.0	6.1	38.2	29692060	4853251.7
Vendor 5	439	4.0	5.8	38.7	29368140	4831630.7
Vendor 4	462	4.0	6.0	38.1	29427301	4821537.1
Vendor 2	402	4.0	6.0	37.3	27832780	4635603.2
Vendor 10	419	4.0	6.0	35.8	27503478	4509427.2
Vendor 7	447	4.0	6.1	41.6	27210565	4487279.4
Vendor 6	408	4.0	5.9	39.0	24809139	4200121.4

- Vendor Ratings are uniform (4.0/5) across all 10 vendors, so rating alone doesn't distinguish top from bottom performers — order volume and delivery metrics vary more meaningfully.
- Vendor 3 leads on order volume and sales (470 orders, Rs 3.09 Cr sales, Rs 50.9L profit), followed closely by Vendor 8 and Vendor 4.
- On-Time Rate varies more than expected (35.8%–41.6%) despite identical ratings — Vendor 7 has the best on-time rate (41.6%) while Vendor 10 has the worst (35.8%), confirming rating isn't a reliable delivery predictor.
- Avg Lead Time is fairly consistent (5.8–6.1 days) across vendors, with Vendor 9 and Vendor 5 slightly faster at 5.8 days.
- Sales and profit scale with order count, not rating — Vendor 6 has the lowest sales/profit (Rs 2.48 Cr / Rs 42L) despite the same 4.0 rating as

the top performers, reinforcing that bulk order volume — not rating — drives cost/profit outcomes.

4. Inventory Status



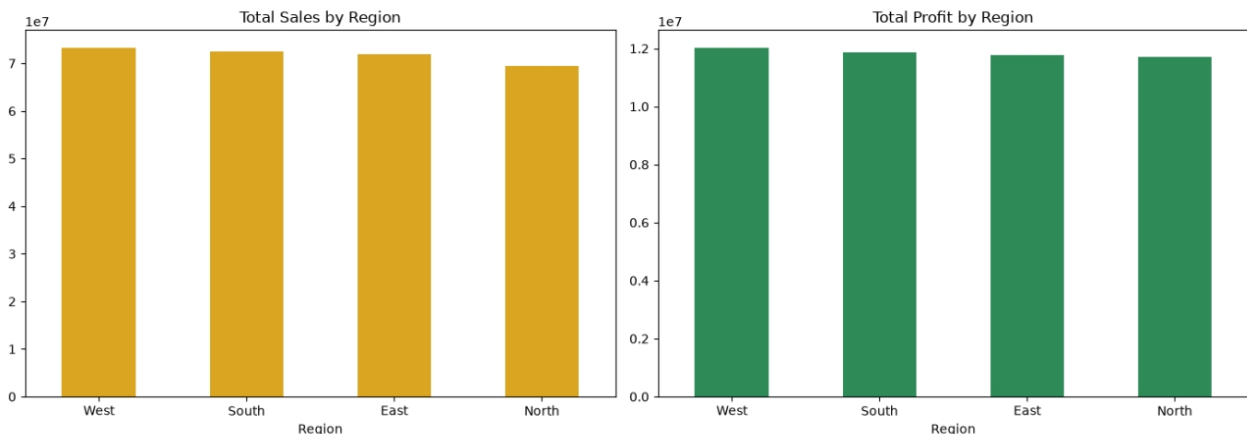
In Stock dominates overwhelmingly: ~3.9K products are In Stock, making up the vast majority of total inventory.

Low Stock is a small fraction: only ~0.5K products fall into the Low Stock category — roughly 11% of the In Stock count.

Out of Stock is essentially zero (0.0K), meaning stockouts are practically nonexistent across the product catalog.

Overall inventory health looks very strong, with minimal restocking risk — the small Low Stock segment is the only area worth periodic monitoring.

5. Total Sales and Profit by Region



- West leads both Sales and Profit, topping ~Rs 7.3 Cr in sales and ~Rs 1.2 Cr in profit, edging out the other three regions.
- South is a close second across both charts, with sales and profit only marginally behind West.
- East and North trail slightly, with North recording the lowest sales (~Rs 6.7 Cr) and lowest profit (~Rs 1.1 Cr) among all regions.
- The gap between regions is narrow overall — no region is dramatically underperforming, suggesting demand and profitability are fairly well distributed geographically.

Final Recommendations

1. Maintain optimal inventory levels by replenishing low-stock products and reducing excess inventory.
2. Improve delivery planning in regions with lower on-time delivery performance.
3. Work closely with low-performing vendors to improve delivery reliability and service quality.
4. Investigate cancelled orders to identify root causes and minimize future cancellations.
5. Optimize warehouse operations by balancing inventory across all warehouse locations.
6. Monitor monthly delivery trends to identify delays and improve logistics planning.